Unit : 5

The Link layer

Data Link Layer

- Data Link Layer is second layer of OSI Layered Model.
- Data link layer is responsible for converting data stream to signals bit by bit and to send that over the underlying hardware.
- At the receiving end, Data link layer picks up data from hardware which are in the form of electrical signals, assembles them in a recognizable frame format, and hands over to upper layer.

Data link layer has two sub-layers:

- **Logical Link Control:** It deals with protocols, flow-control, and error control
- **Media Access Control:** It deals with actual control of media

Functionality of Data-link Layer

Framing

- Data-link layer takes packets from Network Layer and encapsulates them into Frames.Then, it sends each frame bit-by-bit on the hardware. At receiver end, data link layer picks up signals from hardware and assembles them into frames.

Addressing

- Data-link layer provides layer-2 hardware addressing mechanism. Hardware address is assumed to be unique on the link. It is encoded into hardware at the time of manufacturing.

Synchronization

- When data frames are sent on the link, both machines must be synchronized in order to transfer to take place.

Error Control

- Sometimes signals may have encountered problem in transition and the bits are flipped.These errors are detected and attempted to recover actual data bits. It also provides error reporting mechanism to the sender.

Flow Control

- Stations on same link may have different speed or capacity. Data-link layer ensures flow control that enables both machine to exchange data on same speed.

Multi-Access

- When host on the shared link tries to transfer the data, it has a high probability of collision. Data-link layer provides mechanism such as CSMA/CD to equip capability of accessing a shared media among multiple Systems.

Error Detection

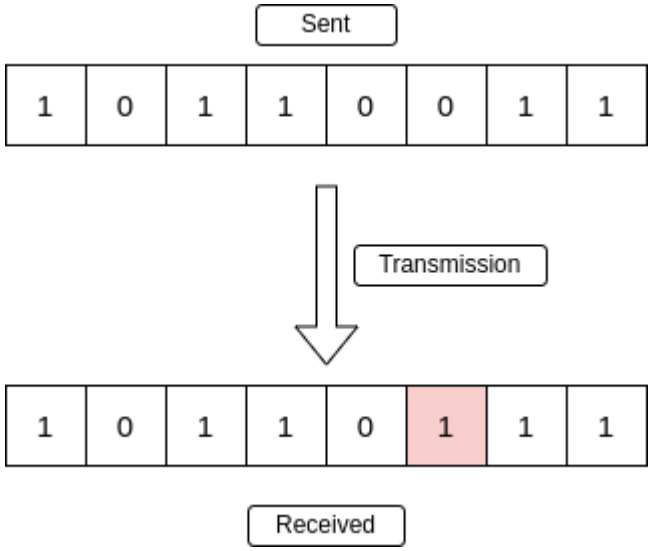
Error detection in computer networks ensures that data is transmitted accurately without corruption. It works by adding extra (redundant) bits to data so the receiver can verify its correctness. If errors are found, the receiver discards the frame and requests retransmission.

- Detects errors caused by noise or interference.
- Uses redundant bits for consistency checking.
- Common methods: Parity Check, Checksum, CRC.
- Ensures data integrity and reliable communication.

Types of Errors

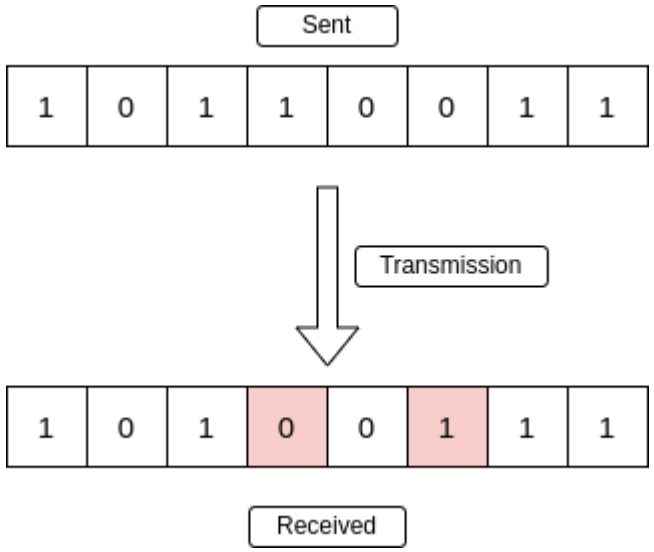
1. Single-Bit Error

- A single-bit error refers to a type of data transmission error that occurs when one bit (i.e., a single binary digit) of a transmitted data unit is altered during transmission, resulting in an incorrect or corrupted data unit.



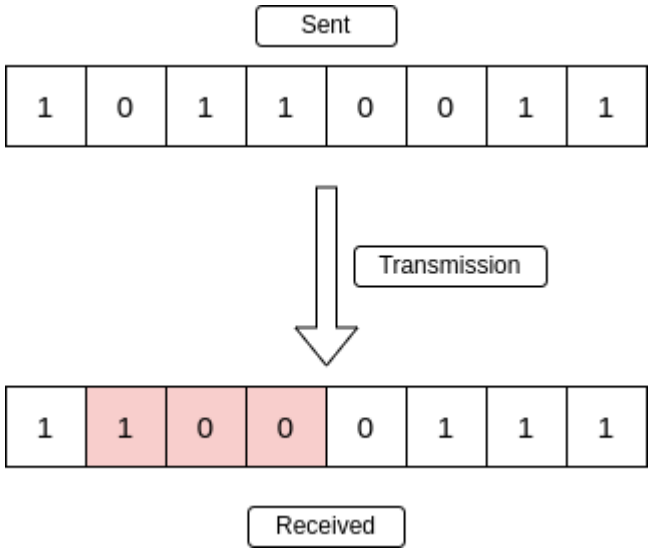
2. Multiple-Bit Error

- A multiple-bit error is an error type that arises when more than one bit in a data transmission is affected.
- Although multiple-bit errors are relatively rare when compared to single-bit errors, they can still occur, particularly in high-noise or high-interference digital environments.



3. Burst Error

When several consecutive bits are flipped mistakenly in digital transmission, it creates a burst error. This error causes a sequence of consecutive incorrect values.



Error Detection Methods

To detect errors, a common technique is to introduce redundancy bits that provide additional information. Various techniques for error detection include:

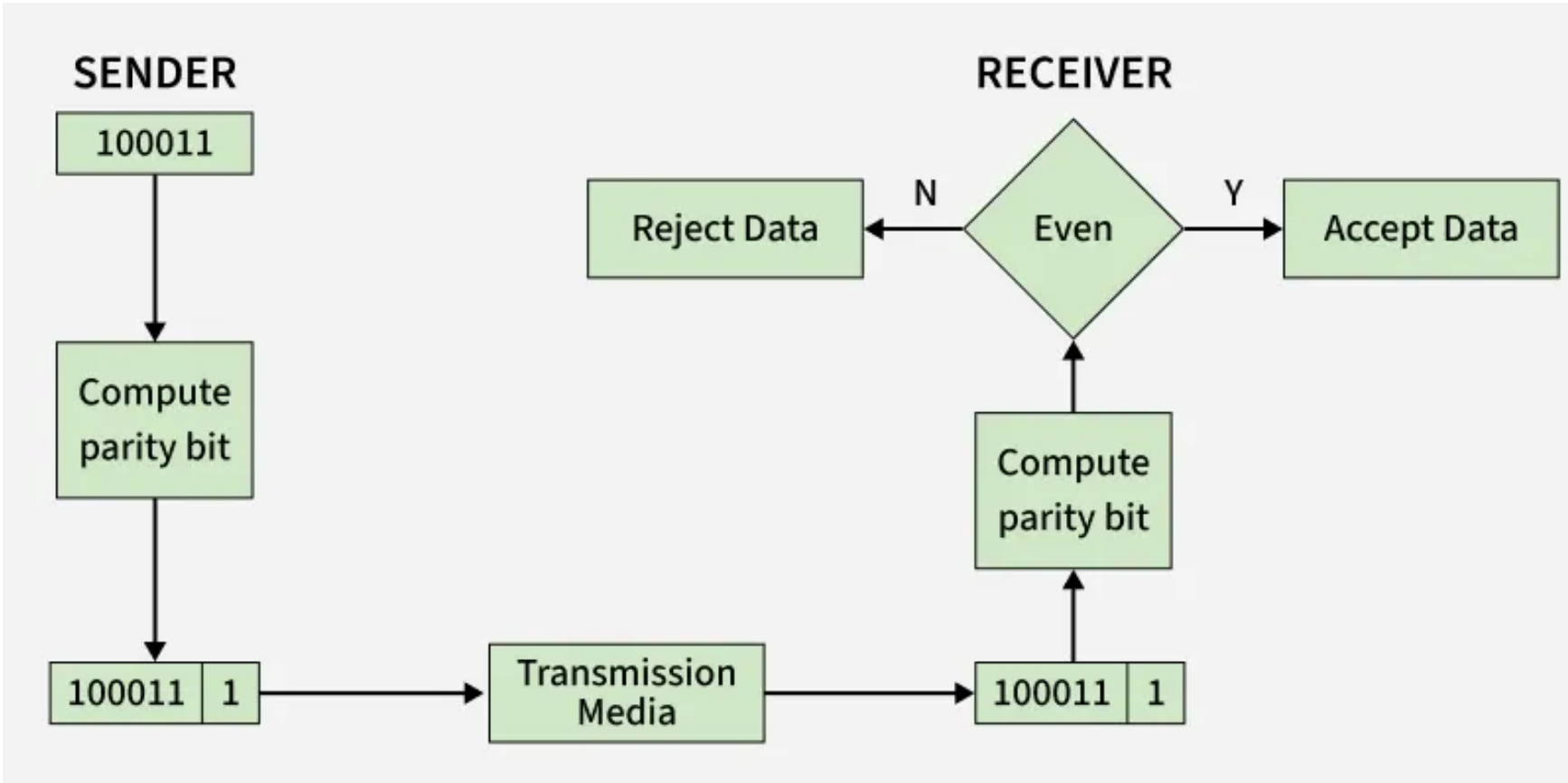
- Simple Parity Check
- Two-Dimensional Parity Check
- Checksum
- Cyclic Redundancy Check (CRC)

1. Simple Parity Check

Simple bit parity is a simple error detection method that involves adding an extra bit to a data transmission. It works as:

- 1 is added to the block if it contains an odd number of 1’s, and
- 0 is added if it contains an even number of 1’s

This scheme makes the total number of 1’s even, that is why it is called even parity checking.


SENDER
100011
Compute parity bit
100011
1
Transmission Media
RECEIVER
100011
1
Compute parity bit
Even
N
Reject Data
Y
Accept Data

Advantages of Simple Parity Check

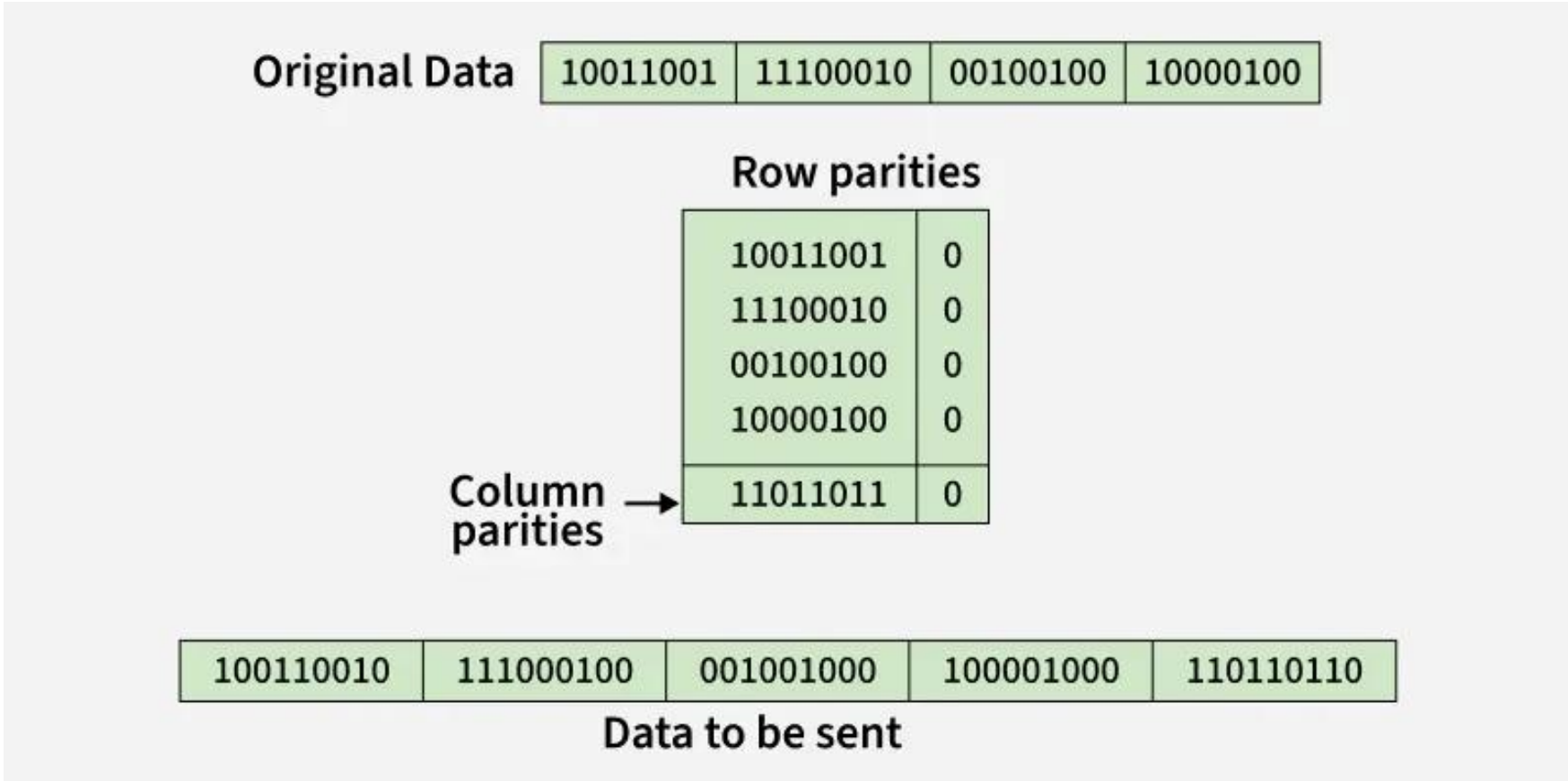
- Simple parity check can detect all single bit error.
- Simple parity check can detect an odd number of errors.
- **Implementation**: Simple Parity Check is easy to implement in both hardware and software.
- **Minimal Extra Data**: Only one additional bit (the parity bit) is added per data unit (e.g., per byte).
- **Fast Error Detection**: The process of calculating and checking the parity bit is quick, which allows for rapid error detection without significant delay in data processing or communication.
- **Single-Bit Error Detection**: It can effectively detect single-bit errors within a data unit, providing a basic level of error detection for relatively low-error environments.

Disadvantages of Simple Parity Check

- Single Parity check is not able to detect even no. of bit error.
- **For example:** the Data to be transmitted is **101010**. Codeword transmitted to the receiver is 1010101 (we have used even parity).
  Let's assume that during transmission, two of the bits of code word flipped to 1111101.
  On receiving the code word, the receiver finds the no. of ones to be even and hence **no error,** which is a wrong assumption.

2. Two-Dimensional Parity Check

- Two-dimensional Parity check bits are calculated for each row, which is equivalent to a simple parity check bit.
- Parity check bits are also calculated for all columns, then both are sent along with the data.
- At the receiving end, these are compared with the parity bits calculated on the received data.


Original Data
10011001
11100010
00100100
10000100
Row parities
10011001 0
11100010 0
00100100 0
10000100 0
Column parities
11011011 0
100110010
111000100
001001000
100001000
110110110
Data to be sent

Advantages of Two-Dimensional Parity Check

- Two-Dimensional Parity Check can detect and correct all single bit error.
- Two-Dimensional Parity Check can detect two or three bit error that occur any where in the matrix.

Disadvantages of Two-Dimensional Parity Check

- Two-Dimensional Parity Check can not correct two or three bit error. It can only detect two or three bit error.
- If we have a error in the parity bit then this scheme will not work.

3. Checksum

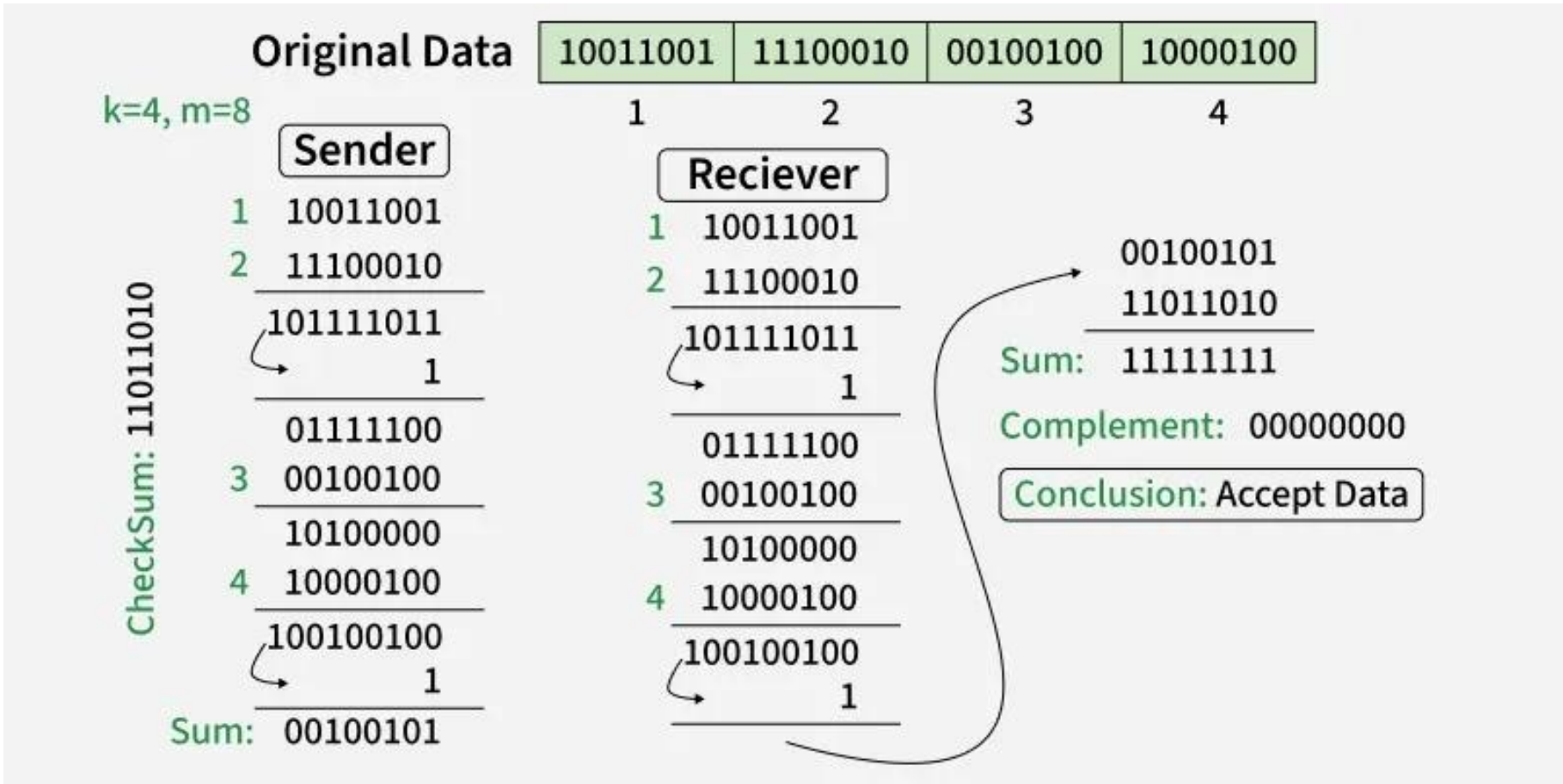
- Checksum error detection is a method used to identify errors in transmitted data.
- The process involves dividing the data into equally sized segments and using a 1’s complement to calculate the sum of these segments.
- The calculated sum is then sent along with the data to the receiver.
- At the receiver's end, the same process is repeated and if all zeroes are obtained in the sum, it means that the data is correct

Checksum - Operation at Sender's Side

- Firstly, the data is divided into k segments each of m bits.
- On the sender’s end, the segments are added using 1’s complement arithmetic to get the sum. The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.

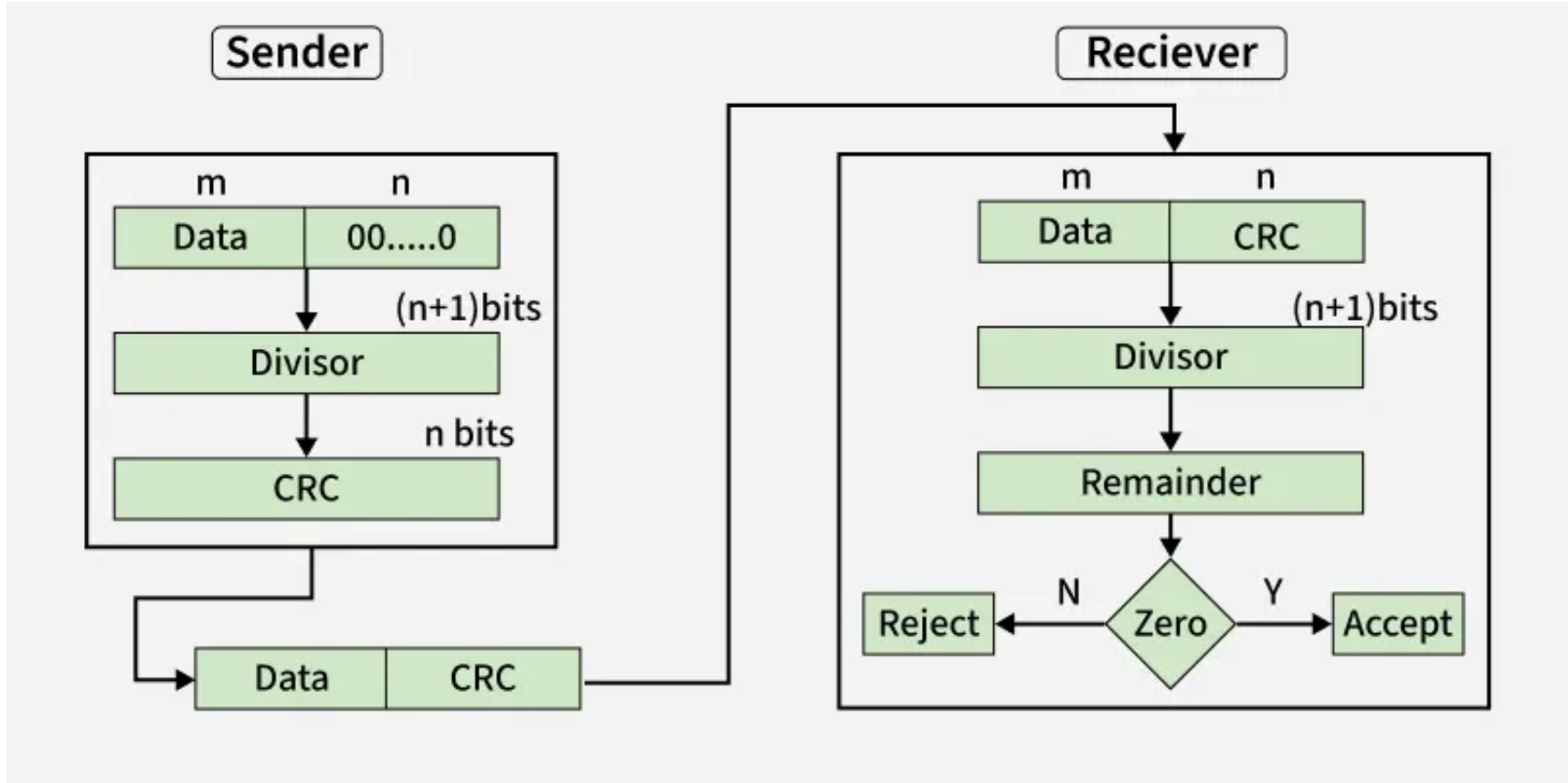
Checksum - Operation at Receiver's Side

- At the receiver’s end, all received segments are added using 1’s complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.


Original Data
10011001
11100010
00100100
10000100
1
2
3
4
k=4, m=8
Sender
1 10011001
2 11100010
101111011
1
01111100
3 00100100
10100000
4 10000100
100100100
1
Sum: 00100101
CheckSum: 11011010
Reciever
1 10011001
2 11100010
101111011
1
01111100
3 00100100
10100000
4 10000100
100100100
1
00100101
11011010
Sum: 11111111
Complement: 00000000
Conclusion: Accept Data

4. Cyclic Redundancy Check (CRC)

- Unlike the checksum scheme, which is based on addition, CRC is based on binary division.
- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of the data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
- At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.


Sender
m
n
Data
00.....0
(n+1)bits
Divisor
n bits
CRC
Data
CRC
Reciever
m
n
Data
CRC
(n+1)bits
Divisor
Remainder
N
Y
Reject
Zero
Accept

CRC Working

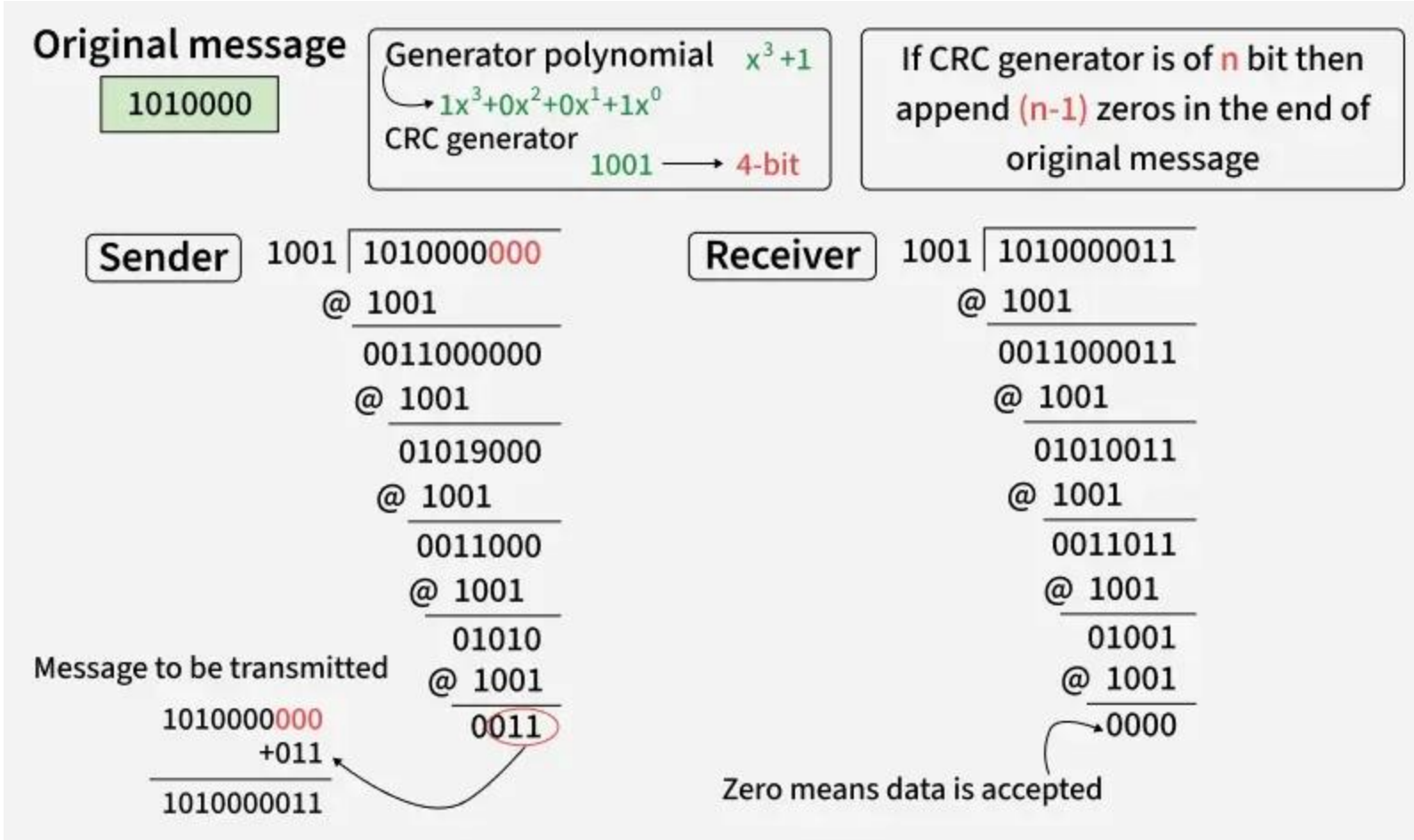
We have given dataword of length n and divisor of length k.

- **Step 1:** Append (k-1) zero's to the original message
  Step 2: Perform modulo 2 division
  Step 3: Remainder of division = CRC
  Step 4: Code word = Data with append k-1 zero's + CRC

Note:

- CRC must be k-1 bits
- Length of Code word = n+k-1 bits

Example: Let's data to be send is 1010000 and divisor in the form of polynomial is x3+1. CRC method discussed below

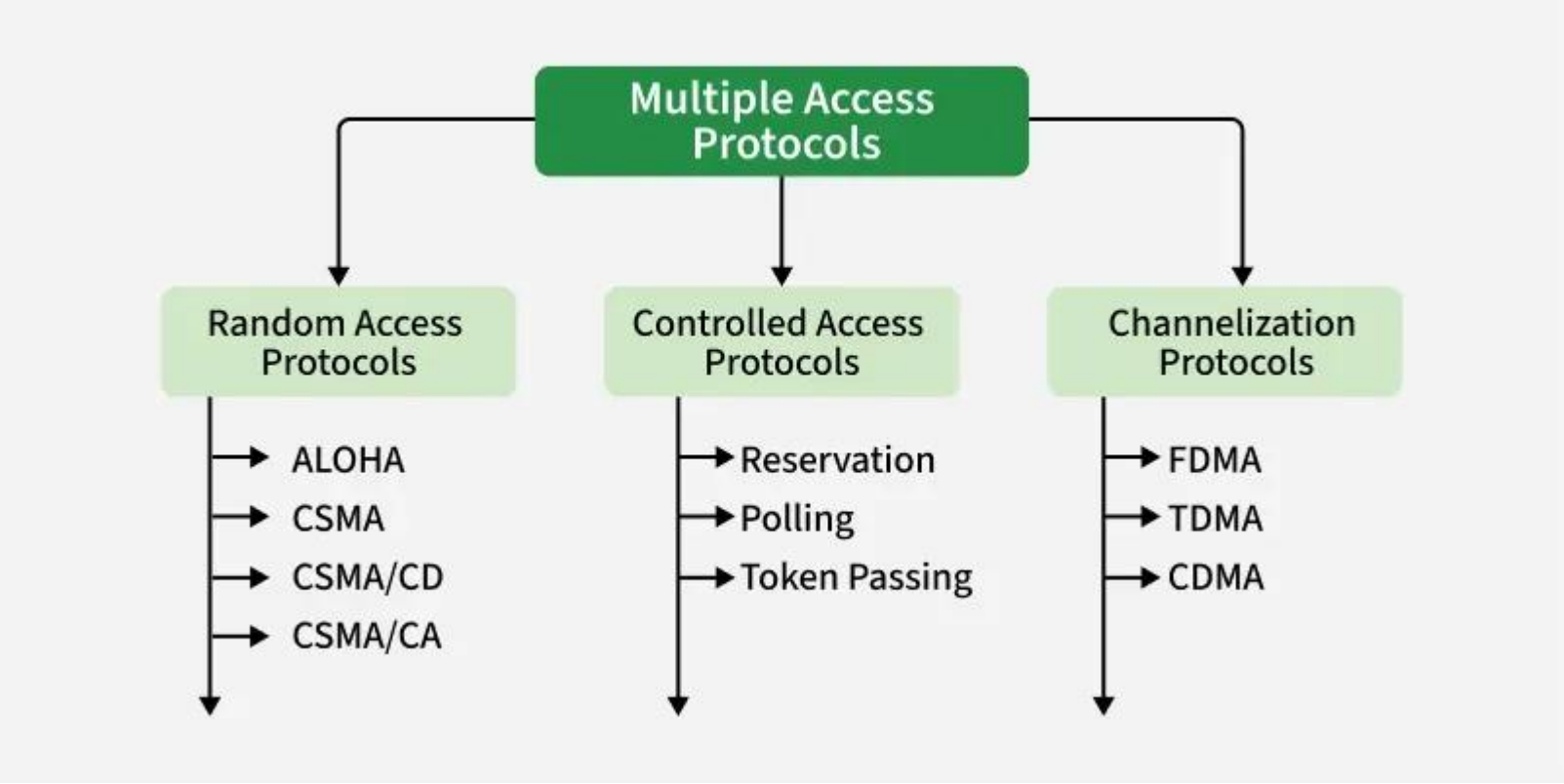


Multiple Access Protocols in Computer Network

- Multiple access protocols are a set of protocols operating in the Medium Access Control sublayer (MAC layer) of the Open Systems Interconnection (**OSI**) model.
- These protocols allow a number of nodes or users to access a shared network channel. Several data streams originating from several nodes are transferred through the multi-point transmission channel.
- The objectives of multiple access protocols are optimization of transmission time, minimization of collisions and avoidance of crosstalks.

Categories of Multiple Access Protocols

Multiple access protocols can be broadly classified into three categories - random access protocols, controlled access protocols and channelization protocols.



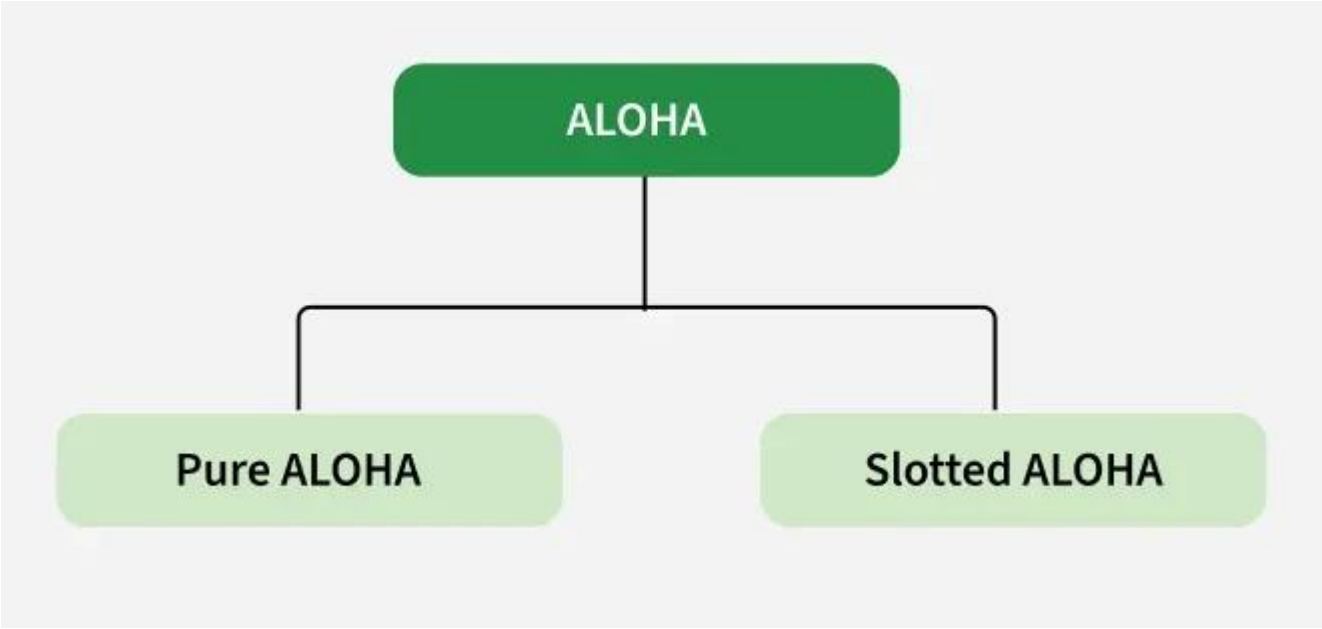
1. Random Access Protocol

In this, all stations have same superiority that is no station has more priority than another station. Any station can send data depending on medium's state(idle or busy). It has two features:

- There is no fixed time for sending data
- There is no fixed sequence of stations sending data

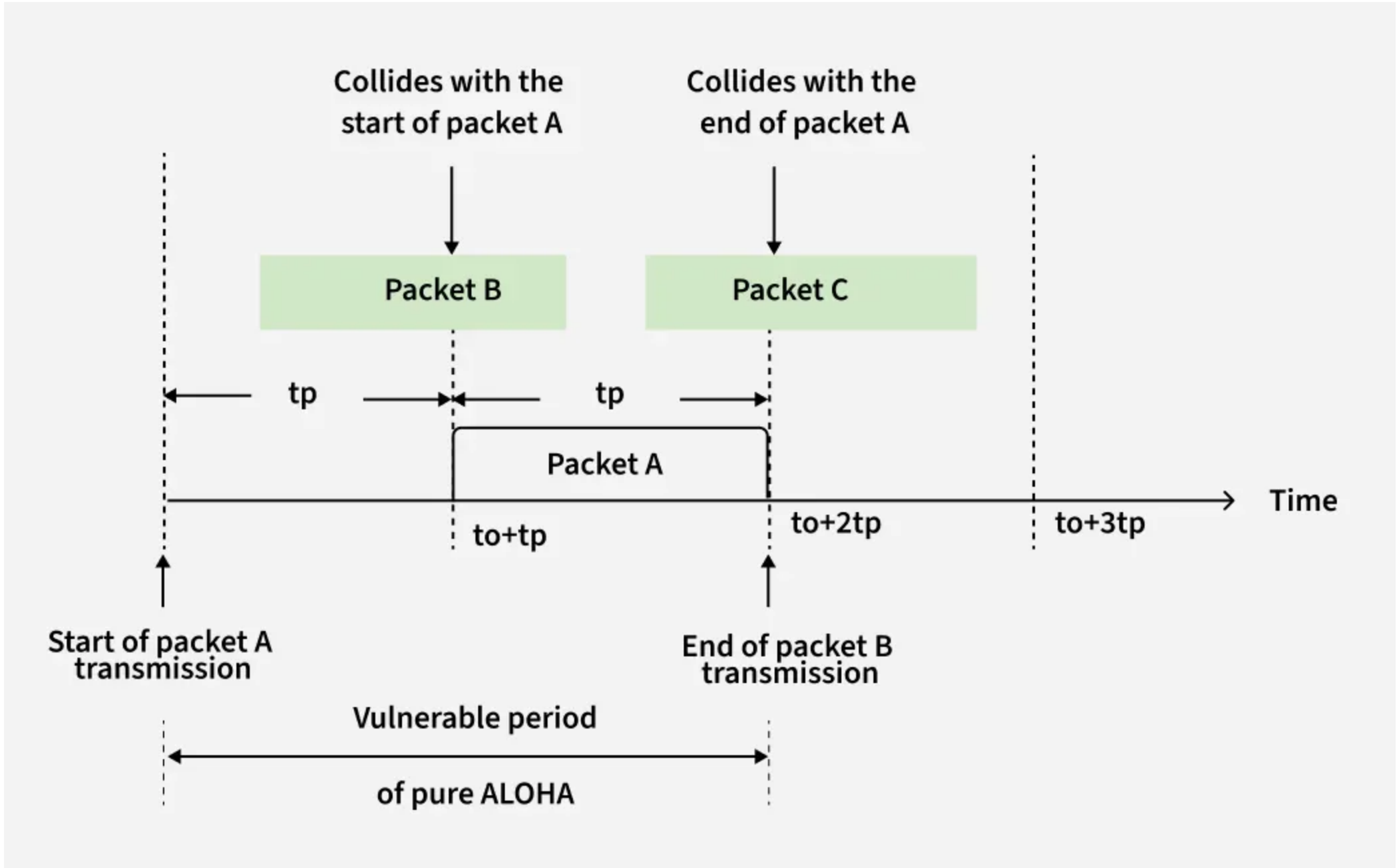
ALOHA

- It was designed for wireless LAN but is also applicable for shared medium.
- In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.



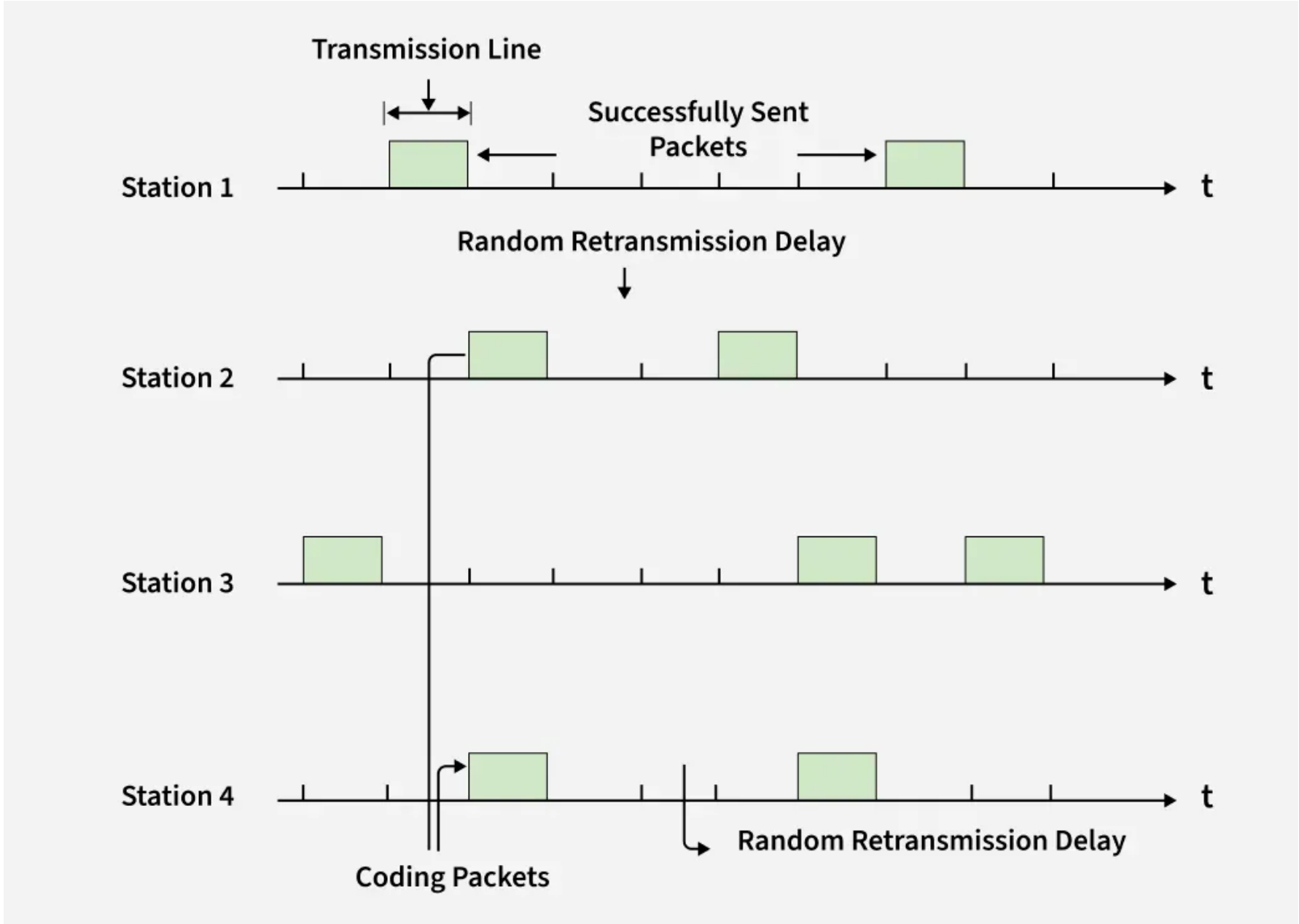
Pure ALOHA

- When a station sends data it waits for an acknowledgement.
- If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time (Tb) and re-sends the data.
- Since different stations wait for different amount of time, the probability of further collision decreases.


Collides with the start of packet A
Collides with the end of packet A
Packet B
Packet C
tp
tp
Packet A
Time
to+tp
to+2tp
to+3tp
Start of packet A transmission
End of packet B transmission
Vulnerable period
of pure ALOHA

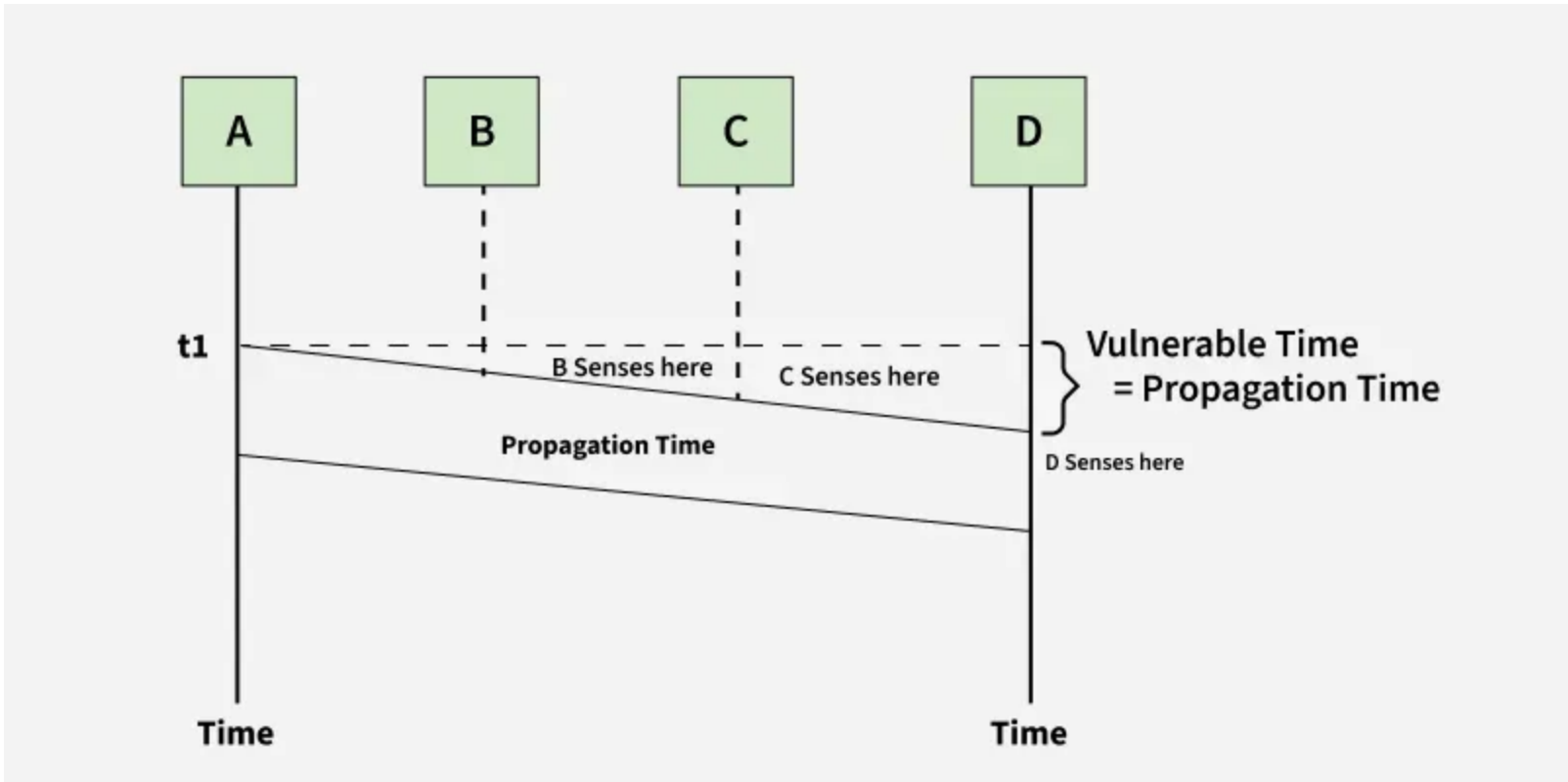
Slotted ALOHA

- It is similar to pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots.
- If a station misses out the allowed time, it must wait for the next slot.
- This reduces the probability of collision.

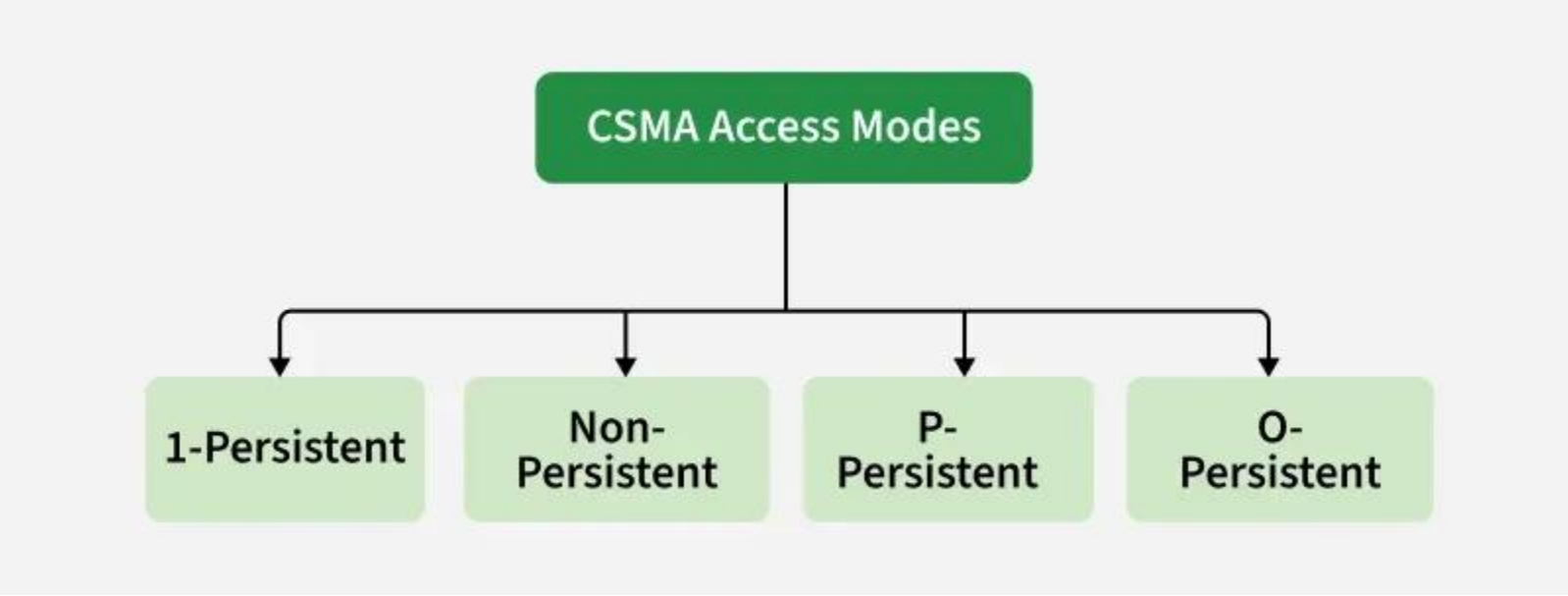

Transmission Line
Successfully Sent
Packets
Station 1
t
Random Retransmission Delay
Station 2
t
Station 3
t
Station 4
t
Random Retransmission Delay
Coding Packets

CSMA

- Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data.
- If it is idle then it sends data, otherwise it waits till the channel becomes idle.
- However there is still chance of collision in CSMA due to propagation delay.
- For example, if station A wants to send data, it will first sense the medium. If it finds the channel idle, it will start sending data. However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data. This will result in collision of data from station A and B.


A
B
C
D
t1
B Senses here
C Senses here
Vulnerable Time
= Propagation Time
Propagation Time
D Senses here
Time
Time

CSMA Access Modes



CSMA/CD

- Carrier sense multiple access with collision detection.
- Stations can terminate transmission of data if collision is detected.

CSMA/CA

- Carrier sense multiple access with collision avoidance.
- The process of collisions detection involves sender receiving acknowledgement signals.
- If there is just one signal(its own) then the data is successfully sent but if there are two signals(its own and the one with which it has collided) then it means a collision has occurred.
- To distinguish between these two cases, collision must have a lot of impact on received signal.

DOCSIS

- Data Over Cable Service Interface Specifications, much better known as DOCSIS is the technology that is used to communicate and transfer data across RF or coaxial cable at high speeds.
- The use of this technology began with DOCSIS 1.0 in 1997 by Cable Labs.
- It was originally established using the pre-existing cable TV infrastructure, making the roll out for this technology possible.
- The standard is now in its third generation, with market prospects for the future remaining bright.

The Evolution of Data Over Cable Service Interface Specification

DOCSIS 1.0

- DOCSIS 1.0 obtained a bandwidth of up to 40 Mbps downstream and 10 Mbps upstream.
- At the time, these rates were sufficient relative to the global usage requirements.
- It didn't take long for people to solidly embrace high-speed internet and within two years DOCSIS 1.1 was released. 1.1 kept the same bit rates (40 Mbps DS/10 Mbps US) but allowed for the implementation of VoIP services for cable TV subscribers.

DOCSIS 2.0

- With the constant increase in demand for more bandwidth, DOCSIS 2.0 was released in 2002.
- A wider band was made available in the upstream channel and modulation schemes were becoming more effective.
- The results of these changes allowed for 40 Mbps download and 30 Mbps upload.

Continued…

DOCSIS 3.0 technology

- In 2006, DOCSIS 3.0 entered the picture, which upped the game significantly.
- This new specification offered channel bonding, providing the ability to combine upstream and downstream channels.
- By connecting 6 or 8 MHz channels, achieves speeds of up to 340 Mb/s (440 Mb/s for EuroDOCSIS downstream and 120 Mbps upstream.
- This release also enabled IP multicast, IPv6 functionality, and AES encryption.

DOCSIS 4.0

- DOCSIS 4.0 is the latest major version of the DOCSIS standard developed by CableLabs to enable very high-speed broadband over existing hybrid fibre-coaxial (HFC) cable networks.
- It builds on earlier versions (DOCSIS 3.0, 3.1) but significantly enhances upstream capacity, spectrum usage, and supports near-symmetric multi-gigabit services.
- The standard was published in 2020.